

1.

Laptop	Price	CPU	Battery	Weight	Storage
A	4500,00	15500,00	6,00	2,00	512,00
B	4000,00	14200,00	7,00	2,20	256,00
C	4800,00	17200,00	5,00	1,80	1024,00
D	4300,00	13800,00	8,00	2,50	512,00
E	4700,00	16000,00	6,00	2,10	512,00
F	4200,00	14500,00	9,00	2,00	256,00

3.

0,633325	0,77687	0,517241	0,393443	0,333333	2,654209
0,126665	0,15537	0,369458	0,393443	0,333333	1,378272
0,090475	0,03107	0,073892	0,147541	0,185185	0,528167
0,079166	0,01942	0,024631	0,04918	0,111111	0,283509
0,070369	0,01726	0,014778	0,016393	0,037037	0,155842
1	1	1	1	1	1

TASK 2

	COST	BENEFIT	BENEFIT	COST	BENEFIT
Laptop	Price	CPU	Battery	Weight	Storage
A	4500,00	15500,00	6,00	2,00	512,00
B	4000,00	14200,00	7,00	2,20	256,00
C	4800,00	17200,00	5,00	1,80	1024,00
D	4300,00	13800,00	8,00	2,50	512,00
E	4700,00	16000,00	6,00	2,10	512,00
F	4200,00	14500,00	9,00	2,00	256,00
MIN	4000,00	13800,00	5,00	1,80	256,00
MAX	4800,00	17200,00	9,00	2,50	1024,00

3. $r=Q^*w$

A	0,44044
B	0,37982
C	0,66764
D	0,25002
E	0,44514
F	0,395

TASK 3

	COST	BENEFIT	BENEFIT	COST	BENEFIT
Laptop	Price	CPU	Battery	Weight	Storage
A	4500,00	15500,00	6,00	2,00	512,00
B	4000,00	14200,00	7,00	2,20	256,00
C	4800,00	17200,00	5,00	1,80	1024,00
D	4300,00	13800,00	8,00	2,50	512,00
E	4700,00	16000,00	6,00	2,10	512,00
F	4200,00	14500,00	9,00	2,00	256,00
MIN	4000,00	13800,00	5,00	1,80	256,00
MAX	4800,00	17200,00	9,00	2,50	1024,00
denominator	10840,2	37341,93	17,05872	5,171073	1402,17
eigenvector	0,27565	0,530842	0,056702	0,031168	0,105633

ranking vector:

A	0,39963
B	0,23526
C	0,75651
D	0,2869
E	0,44146
F	0,24629

TASK 4 COMPARISON

	SAW		TOPSIS	
Laptop	r	ranking	r	ranking
A	0,440441	3	0,39963	3
B	0,379815	5	0,235255	6
C	0,667644	1	0,756512	1
D	0,250022	6	0,286897	4
E	0,44514	2	0,441465	2
F	0,394997	4	0,246289	5

2.

	CPU	Price	Storage	Battery	Weight
CPU		1,00	5,00	7,00	8,00
Price		0,20	1,00	5,00	8,00
Storage		0,14	0,20	1,00	3,00
Battery		0,13	0,13	0,33	1,00
Weight		0,11	0,11	0,20	0,33
SUM		1,58	6,44	13,53	20,33

4.eigenvector

PRICE	0,275654481	2			
CPU	0,530841786	1	0,275654481	0,530841786	0,056702
BATTERY	0,056701853	4			0,031168
WEIGHT	0,031168389	5			0,105633
STORAGE	0,105633491	3			
	1				

2. qj table

in each column should be at least one 0 and 1 (passed)					
0,375	0,500	0,250	0,714	0,333333333	
1,000	0,118	0,500	0,429	0	
0,000	1,000	0,000	1,000	1	
0,625	0,000	0,750	0,000	0,333333333	
0,125	0,647	0,250	0,571	0,333333333	
0,750	0,206	1,000	0,714	0	

qij table	COST	BENEFIT	BENEFIT	COST	BENEFIT	S*	S-
	0,114430	0,220343	0,019944	0,012055	0,038572	0,048315	0,032161
	0,101716	0,201863	0,023267	0,013260	0,019286	0,072224	0,022218
	0,122059	0,244510	0,016620	0,010849	0,077144	0,024303	0,075508
	0,109344	0,196177	0,026591	0,015069	0,038572	0,062538	0,02516
	0,119516	0,227451	0,019944	0,012658	0,038572	0,046887	0,037059
	0,106801	0,206128	0,029915	0,012055	0,019286	0,069628	0,022752
a*	0,101716	0,244510	0,029915	0,010849	0,077144		
a-	0,122059	0,196177	0,016620	0,015069	0,019286		

RESULTS:	The top 3 positions are exactly the same in both methods.	The eigenvector is exactly the same for both methods
	The bottom 3 positions contain the same laptops, however they swap places between the methods.	
	Laptop C is the best in both methods because the most important criterion (CPU: 53%) reaches its highest value for C, which is also the lightest and has the largest storage. Even though C has the worst battery, the other factors outweigh it.	

possible reasons:	SAW assigns a value of 0 to the worst element in each column due to min-max scaling. TOPSIS, on the other hand, penalizes quadratically (via Euclidean distance), but no value is forced to 0
	TOPSIS tends to favor balanced options while SAW penalizes alternatives heavily for being the worst in a specific criterion. For example, laptop D is balanced but has the worst CPU, which is the reason it ranks last in the SAW method, while in TOPSIS it climbs up to 4th place.